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Rural distribution system reliability worth evaluation using individual customer outage cost characteristics

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Abstract

Customer interruption costs are a function of both interruption and user characteristics. The specific customer characteristics at a load point in a distribution network can have a significant effect on the estimated customer outage costs at that load point and for the entire feeder. This paper illustrates the effects on outage cost estimates of using generalized sector customer outage cost estimates and actual customer specific outage cost functions. The data used in these analyses are from customer cost surveys conducted in Canada. A small distribution system is used to illustrate the possible variation in interruption cost estimates due to incorporating specific customer characteristics in the evaluation. The results presented clearly show that different customer compositions can result in significantly different feeder interruption cost estimates. The estimated interruption costs can be used in feeder design to create an optimal configuration based on reliability cost/worth analysis.
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1. Introduction

Considerable work has been done on the development of quantitative techniques for distribution system reliability evaluation [1]. The basic reliability indices in a distribution system predictive assessment are the frequency of failure or rate of failure (λ), the average duration of failure (r) and the average annual outage time (U) at each load point. These basic predictive indices can be combined with load point customer data to obtain the system average interruption frequency index (SAIFI), the system average interruption duration index (SAIDI), the customer average interruption duration index (CAIDI) and the average service availability index (ASAI), parameters normally associated with past performance assessment. The SAIFI, SAIDI, CAIDI and ASAI parameters can be calculated for a single feeder, a system area or for the entire system. It is relatively simple [1] to extend the predictive process to incorporate customer outage costs in the evaluation and assess the reliability worth associated with parameter or system design changes.

Customer interruption costs can be used as a practical surrogate for reliability worth [2]. The most practical way to obtain realistic estimates of the monetary damage incurred by customers due to electric power supply failures is to specifically contact customers and ask for this information. The Power System Research Group at the University of Saskatchewan has conducted a number of surveys to evaluate residential, commercial, industrial, farm, government and institutional customer outage costs. The first surveys were conducted in 1981. The data shown in this paper were obtained from the surveys conducted in 1992 [3]. Customer interruption costs are a function of both interruption and user characteristics. The interruption costs can be presented as a function of outage duration and when expressed in this form are known as customer damage functions (CDF) [2]. The CDF can be determined for individual consumers, customers groups or for specific customer sectors. Appropriate CDF can be combined at a load point to produce representative load point CDF (LPCDF) which can be used to estimate the customer outage costs at each load point in a distribution system [1].

The most common customer interruption cost representation is in the form of sector CDF designated as SCDF. SCDF are developed by weighting the respective individual CDF (ICDF) in that sector by their fraction of group energy utilization. A similar procedure is used to create a LPCDF using either ICDF or SCDF. Fig. 1 shows the agricultural SCDF obtained from the 1987 and 1992 Canadian surveys.

The SCDF are extremely useful and provide a reasonable representation of the average outage costs for the various customer sectors. They are, however, representative CDF and can be quite inaccurate when applied in situations in which the load point supplies customers whose ICDF are

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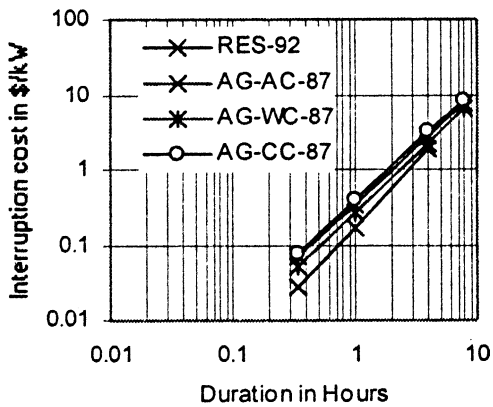

Fig. 1. SCDF for agricultural sector (1982) and residential sector (1992). RES: Residential data (1992); AG-AC: Agricultural All Canada data (1987); AG-WC: Agricultural Western Canada data (1987); AG-CC: Agricultural Central Canada data (1987).

very different from the SCDF. This situation is illustrated in this paper using a simple distribution system configuration and data from the 1987 survey [4] for the agricultural sector and 1992 data for the residential sector.

2. Reliability and system indices

The estimated customer interruption costs using both ICDF and SCDF are presented using a segment of a small test system designated as the RBTS. The RBTS was developed at the University of Saskatchewan and has been used extensively in recent years in the reliability assessment of generation, transmission, distribution and composite systems. This system includes a consistent and complete set of data that enables a wide range of techniques and applications to be easily compared. Bus 6 in the RBTS is

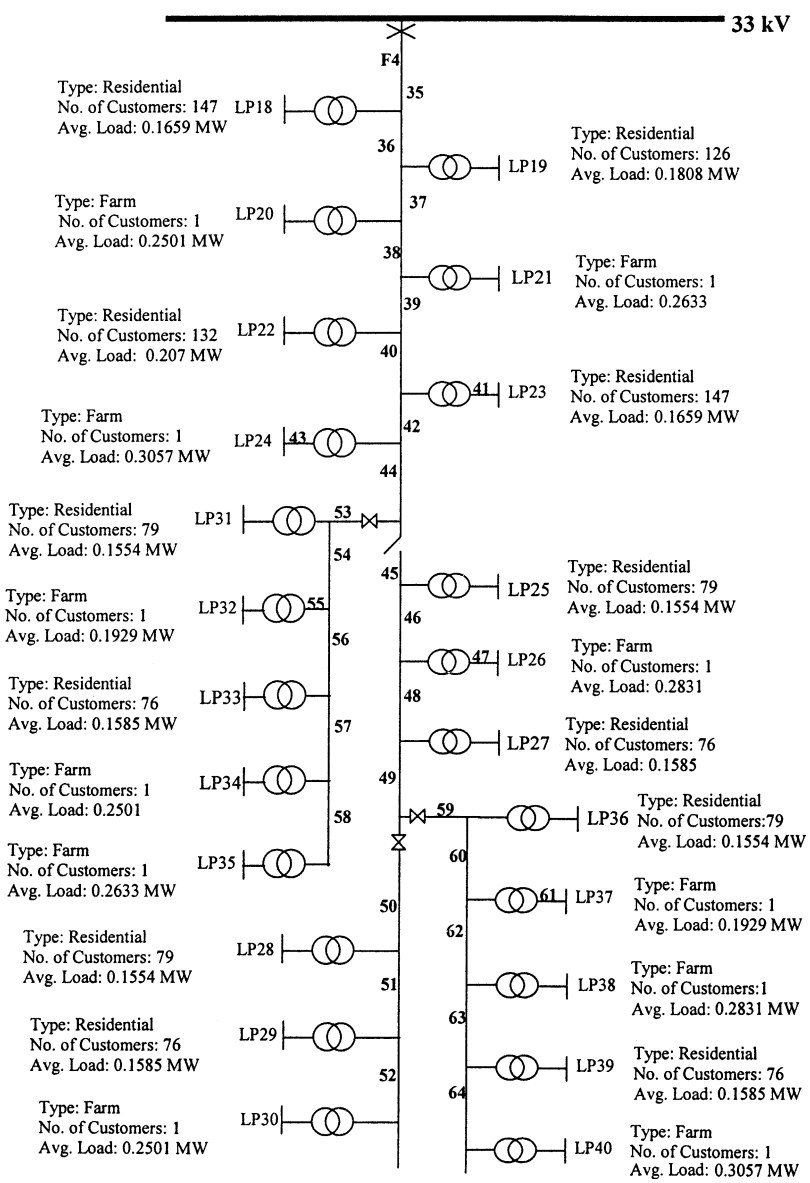

Fig. 2. Feeder 4 at Bus 6 in the RBTS.

Table 1
Feeder 4 base case load point indices

LP	L_{avg} (kW)	n cust	λ (f/yr)	r (h)	U (h/yr)	EENS (kWh/yr)	ECOST ($/yr)
18	165.9	147	1.1880	3.34	3.97	659.1	290.75
19	180.8	126	1.1880	3.34	3.97	718.2	316.86
20	250.1	1	1.1880	3.34	3.97	993.5	733.01
21	263.3	1	1.1880	3.34	3.97	1046.0	771.70
22	207	132	1.1880	3.34	3.97	822.3	362.78
23	165.9	147	1.2156	3.38	4.11	681.9	301.57
24	305.7	1	1.2225	3.39	4.15	1267.2	937.38
25	155.4	79	1.1880	5.06	6.02	934.7	446.91
26	283.1	1	1.2156	5.06	6.15	1741.9	1378.89
27	158.5	76	1.1880	5.06	6.02	953.4	455.83
28	155.4	79	1.5790	5.05	7.97	1238.5	590.47
29	158.5	76	1.5790	5.05	7.97	1263.2	602.24
30	250.1	1	1.5790	5.05	7.97	1993.3	1575.05
31	155.4	79	1.7998	3.91	7.03	1092.7	496.97
32	192.9	1	1.8366	3.93	7.22	1391.9	1056.67
33	158.5	76	1.7998	3.91	7.03	1114.5	506.88
34	250.1	1	1.7998	3.91	7.03	1758.6	1333.85
35	263.3	1	1.7998	3.91	7.03	1851.6	1404.25
36	155.4	79	1.7814	5.04	8.98	1395.8	664.78
37	192.9	1	1.8159	5.04	9.15	1765.9	1394.27
38	283.1	1	1.7814	5.04	8.98	2542.8	2007.87
39	158.5	76	1.7814	5.04	8.98	1423.6	678.04
40	305.7	1	1.7814	5.04	8.98	2745.8	2168.16

a typical load point with a combination of residential, commercial, industrial and agricultural customers. The distribution system at Bus 6 contains four feeders, 40 load points, 42 main sections and 40 lateral sections. There is a circuit breaker located at the beginning of each feeder. Feeder 4, which is composed of farm customers, has been used in the analysis described in this paper. Fig. 2 shows the single line diagram of Feeder 4 including the number and type of customers, and their average loads.

Tables 1 and 2 present the results of a basic reliability analysis of Feeder 4 shown in Fig. 1. The relevant circuit reliability data are given in Ref. [5]. The expected interruption cost (ECOST) values for each load point (LP) in Table 1 were determined using the SCDF shown in Fig. 1. The total feeder ECOST is shown in Table 2.

Feeder 4 is a radial configuration consisting of residential and agricultural consumers. Table 2 shows the overall feeder reliability. The customers at load point 18 are residential consumers and have a relatively high reliability due to their location. They also have low outage costs due to their location, energy utilization and their SCDF.

The total feeder ECOST, as shown in Table 2, is $20,475. This value was obtained using the residential SCDF shown in Fig. 1. The SCDF are representative values, which aggregate the various ICDF in the sector. The SCDF are useful data in general situations where detailed customer information is unavailable. They can, however, misrepresent specific situations containing customers with quite different ICDF, as illustrated in Section 3.

3. Individual customer damage functions

3.1. Residential sector

Interruption cost is a function of interruption duration and user characteristics. The interruption cost associated with a given power supply interruption depends on the attributes of the load point under consideration. For example, the interruption cost at a single customer load point is completely dependent on the cost characteristics of that particular customer. In the case of a multi-customer load point, the resulting cost function represents the combined costs of all customers at that load point. The survey responses from the residential customers were aggregated to develop the residential sector customer damage function shown in Fig. 1. These data are presented numerically in Table 3. For the residential sector, the ICDF and SCDF are the same as there are no sub-sectors embedded in the residential sector.

3.2. Agricultural sector LPCDF

The agricultural customer responses were grouped into various SIC groups to create the ICDF shown in Figs. 3–5. In this sector, the cost data for 27 representative SIC sub-groups were used to create the various ICDF. The range of interruption costs was used to group the various SICs into the three figures based on their 8-h interruption costs. Fig. 3 shows those SIC which have values up to 7 $/kW. Fig. 4 shows those SIC for the 7–10 $/kW range and Fig. 5 presents those with 8-h costs above 10 $/kW. These figures clearly show the range and shapes of the respective ICDF.

The differences in user outage costs are clearly reflected in the ICDF for each sub-group. Each ICDF provides a monetary estimate of how a power failure affects the specific user group. For example, the greenhouse customer (SIC 162 in Fig. 4) is affected in a significantly different manner than an oilseed producer (SIC 133 in Fig. 5). Residential customers experience quite different problems due to power supply failures than do agricultural users and therefore there is not necessarily any similarity in the outage

Table 2
Feeder 4 base case system indices

SAIFI (Int/cst/yr)	SAIDI (h/cst/yr)	CAIDI (h/cst/yr)	EENS (kWh/yr)	AENS (kWh/cst int)	ASAI	ECOST ($/yr)
1.404068	5.864374	4.18	31396	26.54	0.999331	20.475

Table 3
Residential SCDF

Outage duration (h)	SCDF ($/kW)
0.33	0.0278
1.00	0.1626
4.00	1.8126

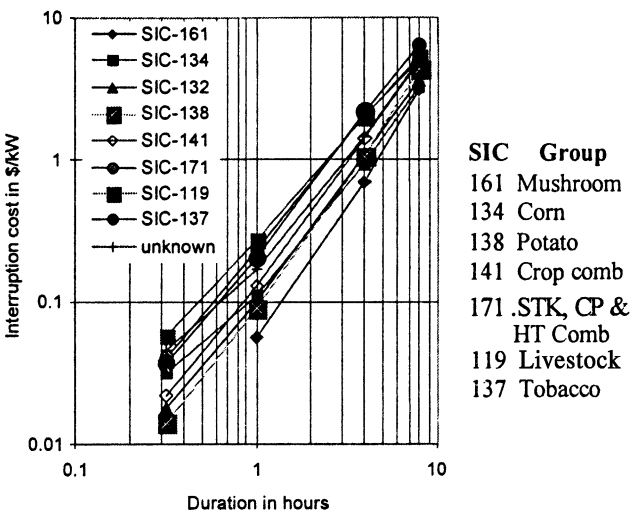

Fig. 3. Agricultural ICDF (Group I).

costs between the SIC sub-groups within a sector or between the sectors.

4. Load point customer damage functions

The LPCDF for those load points which only have residential customers are the same as their ICDF as there are no other sub-groups in the residential sector. The expected cost for the residential consumers, therefore, remains unchanged in the different customer compositions considered for this feeder. The All Canada cost data was used

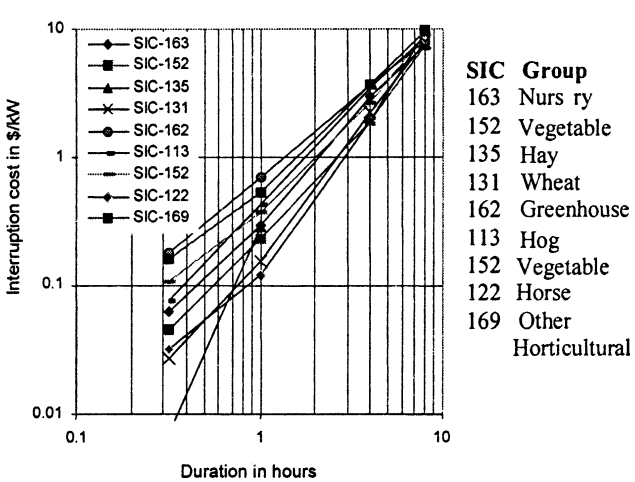

Fig. 4. Agricultural ICDF (Group II).

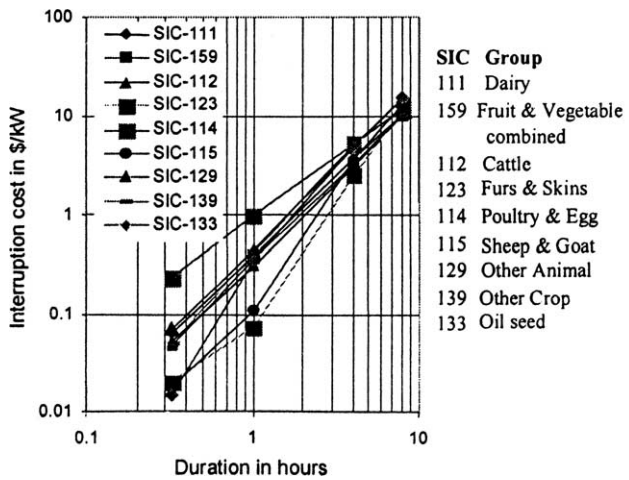

Fig. 5. Agricultural ICDF (Group III).

[5] to establish the individual agricultural CDF. The LPCDF for the selected agricultural customers used in this paper are shown in Table 4.

Four different cases (I, II, III and IV) using different combinations of agricultural customers were created to illustrate the sensitivity of the total feeder expected customer costs. Extrapolation and interpolation were used to obtain LPCDF for the event outage durations encountered in the load point failure mode and effect analysis [1]. These values are shown in Table 5. The case descriptions are as follows:

- Case I has only small grain customers at all farm load points
- Case II has only Cattle Customers at all farm load points
- Case III has 60% Cattle + 40% small grain customers at all farm load points
- Case IV has Hog Customers at LP 20, 30, 34 and 40, and the remaining load points have small grain customers

In Case I, farmers cultivate only small grain at all the agricultural load points. Small grain farmers have the lowest customer damage functions of all the four cases. Raising and breeding cattle is relatively expensive and the farmers make

Table 4
LPCDF for the agricultural customers

Outage duration	LPCDF ($/kW)			
Hours	Small grains	Cattle	Cattle (60%) + Small grain (40%)	Hog
0.330	0.018	0.053	0.039	0.077
1.000	0.099	0.310	0.226	0.430
4.000	1.100	3.050	2.270	3.420
8.000	3.570	10.380	7.670	8.670

Table 5
LPCDF for the required outage event durations

Outage duration	LPCDF ($/kW)			
Hours	Case I	Case II	Case III	Case IV
1	0.099	0.310	0.2256	0.430
5	1.607	4.526	3.360	4.614
10	4.879	14.283	10.520	11.374

significant use of electrical appliances for this purpose. Case II, therefore, has the highest customer damage function. Case III is a mixture of small grain (40%) and cattle (60%) farmers. Case IV is a mixture of hog and grain farmers. The LPCDF shown in Table 4 for the agricultural customers and the SCDF in Table 3 for the residential customers were used at the respective load points in the system.

5. Ecost variability in feeder 4

This section presents the results of the sensitivity study of the ECOST to changes in the customer combinations on Feeder 4. The ECOST for the different agricultural customer combinations considered using the LPCDF in Table 5 are presented in Table 6 for each load point.

It can clearly be seen from Table 6 that it is important to specify the customer types located at the various load points in order to obtain an accurate assessment of the individual load point ECOST and that the overall feeder ECOST can vary considerably with customer composition. Table 7 presents the overall feeder ECOST for the different combinations considered.

Table 6
Results for Feeder 4

ECOST ($/yr)					
Load point	Type	Case I	Case II	Case III	Case IV
18	Residential	290.75	290.75	290.75	290.75
19	Residential	316.86	316.86	316.86	316.86
20	Farm	297.15	842.89	624.80	862.03
21	Farm	312.84	887.39	657.78	312.83
22	Residential	362.79	362.79	362.78	362.79
23	Residential	301.57	301.57	301.57	301.57
24	Farm	380.16	1078.00	799.13	380.16
25	Residential	446.91	446.91	446.91	446.91
26	Farm	566.89	1598.82	1186.47	566.88
27	Residential	455.83	455.83	455.83	455.83
28	Residential	590.47	590.47	590.47	590.47
29	Residential	602.25	602.25	602.25	602.25
30	Farm	646.85	1823.76	1353.48	1847.62
31	Residential	496.97	496.97	496.97	496.97
32	Farm	430.24	1216.32	902.20	430.24
33	Residential	506.88	506.88	506.88	506.88
34	Farm	543.03	1535.35	1138.81	1568.08
35	Farm	571.69	1616.38	1198.92	571.69
36	Residential	664.78	664.78	664.78	664.78
37	Farm	572.34	1613.45	1197.44	572.34
38	Farm	824.28	2323.71	1724.55	2355.81
39	Residential	678.04	678.04	678.04	678.04
40	Farm	890.08	2509.22	1862.23	890.08

Table 7
Feeder 4 ECOST

Case	ECOST ($/yr)
I	11,749.91
II	22,759.36
III	18,359.86
IV	16,071.85

It is important to appreciate that the system reliability indices SAIFI, SAIDI, CAIDI, EENS, AENS, and ASAI are the same for each of the considered customer combinations. These values are shown in Table 2.

Tables 1 and 2 show the individual load point and total feeder ECOST, respectively, when the SCDF shown in Fig. 1 are applied. The total feeder ECOST is $20,475/yr in this case. This can be contrasted with the values shown in Table 7. The actual customer identity is lost when the SCDF are applied to all customers within a sector. Tables 6 and 7 clearly show that the ECOST depends on the load point compositions, in addition to the respective energy utilization, at the various load points. Case I has the lowest ECOST due to the relatively low ICDF associated with grain farmers. Case II results in the highest ECOST due to the cattle farmer sub-groups used in this case. Cases III and IV further illustrate the variability in the ECOST due to type and location of different agricultural electrical energy consumers.

6. Conclusions

SCDF provide a very useful representation of the interruption costs associated with a general group of customers. There are, however, customer sub-groups within each sector that have quite different CDF. The CDF used at a particular load point should reflect as much as possible the actual consumer characteristics at that load point. In many cases, this cannot be accomplished using SCDF. This is illustrated in this paper using a relatively simple distribution system and a set of CDF obtained from customer surveys in Canada. The results presented in the paper clearly show that different customer compositions can result in significantly different feeder interruption cost estimates. The estimated interruption costs can be used in feeder design to create an optimal configuration based on reliability cost/worth analysis.

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